InHand Networks

High-speed 5G, Wi-Fi 6, Cloud-managed

VR624 Vehicle Router

- 5G
- Wi-Fi 6
- GNSS
- M12

InHand Networks

1. Product Overview

The VR624 is a robust 5G router engineered for in-vehicle networking, featuring 4×M12 X-code Gigabit Ethernet ports, RS232 serial, and DC 9-48 V input with reverse-polarity protection. It delivers 5G NR SA/NSA speeds up to 3.4 Gbps, dual Nano-SIM with eSIM optional, Wi-Fi 6 3000 Mbps, and multi-constellation GNSS. Operating from -30°C to +70°C fanless cooling, the VR624 is certified to CE, E-Mark (ECE R10/ECE R118), FCC, PTCRB, and ITxPT standards. With TPM 2.0 hardware encryption and M12 vibration-resistant connectors, the device ensures secure, reliable connectivity for transportation fleets. Compatible with DeviceLive for centralized fleet management and remote diagnostics.

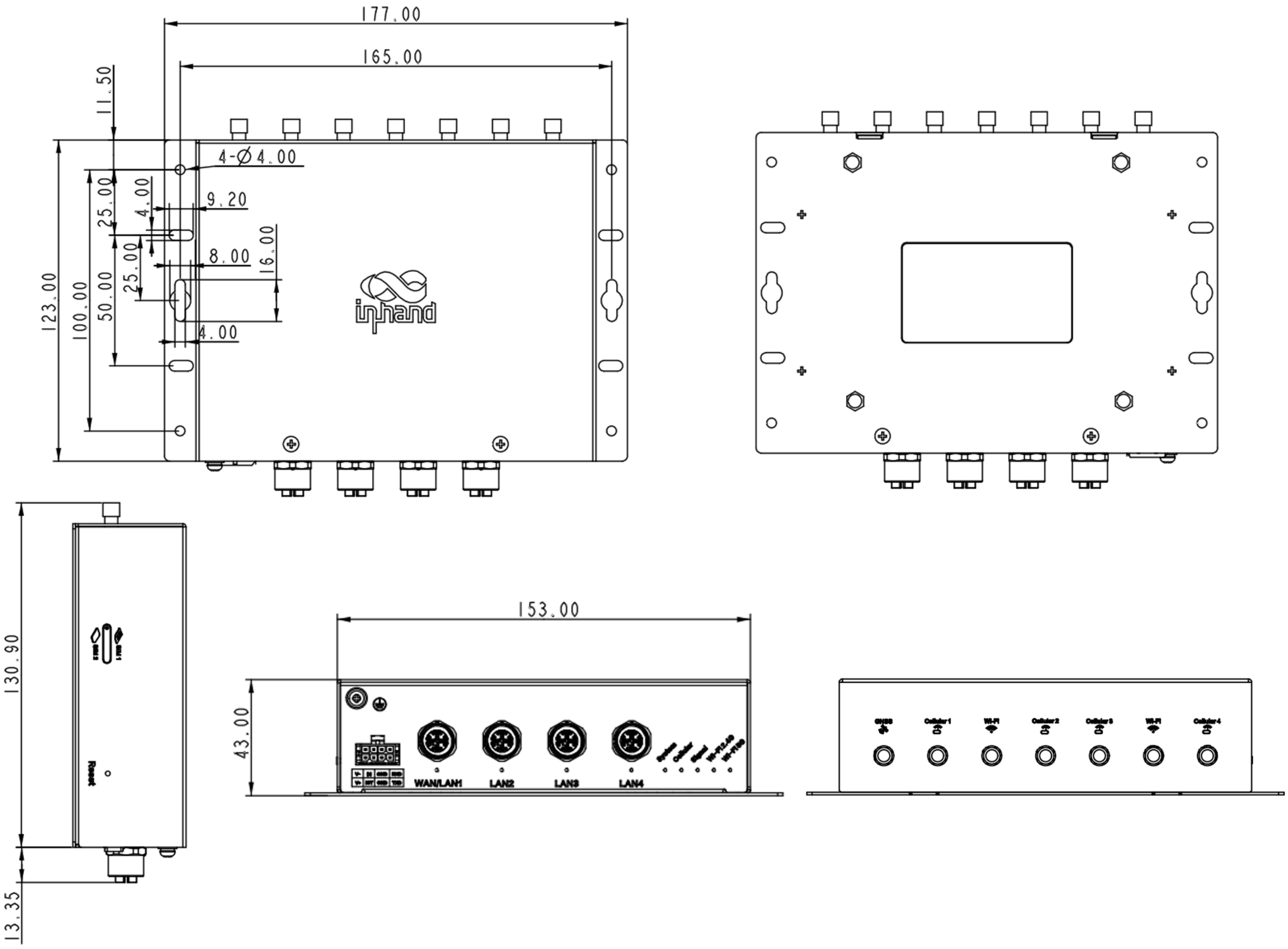
1.1 Feature and Advantage

Value Dimension	Description
High-Speed Connectivity	5G NSA/SA support with downlink speeds up to 3.4 Gbps, Wi-Fi 6 throughput 3000Mbps, enabling gigabit-class data transmissionfor real-time video, telemetry, and fleet communications
Robust Design	M12 X-Coded connectors with vibration resistance, certified to ECE R10, ECE R118, and EN 50155 standards for EMC compliance, fire safety, and railway-grade reliability
Integrated Positioning	Built-in GNSS support (GPS, BDS, Galileo, GLONASS, QZSS) with <1.5m positioning accuracy for real-time fleet tracking and route optimization
Enterprise-Grade Security	Hardware TPM 2.0 encryption chip combined with multi-layer software security policies (firewall, VPN, access control) for comprehensive threat protection, WPA3 Enterprise level secure encryption
Cloud-Managed	Integrated with InHand DeviceLive platform for real-time monitoring, remote diagnostics, and centralized fleet management
ITxPT Compliance	Certified to ITxPT standards, ensuring seamless integration with ITS/ITCS solutions for public transportation systems

1.2 Application Scenarios

Scenario	Typical Customers	Key Requirements	Value Proposition
Public Transit	Bus operators, intercity coach companies	Gigabit Ethernet, vehicle tracking, surveillance backhaul, Passager Wi-Fi	High-speed connectivity for Ticket vending machines, vehicle mounted MDVR etc.,
Emergency Vehicles	Police, fire, ambulance fleets	Real-time data transmission, mission-critical reliability	Ultra-low latency for emergency communications and telemedicine
RV Rental	Motorhome, Campervan	Passenger PC, smartphone connect Internet	Wi-Fi 6 supports simultaneous high-bandwidth usage for streaming, video calls, and remote work
Unmanned Vehicle	RoboShuttle, RoboSweep	Route optimization, real-time tracking	Customer satisfaction 5G connectivity for efficient fleet operations

1.3 Product Dimensions



Note:

1. All dimensions are in millimeters (mm).
2. Dimensions (L × W × H): 177 × 144.25 × 43 mm.
3. All dimensions are approximate, for reference only.
4. Dimensions shown shall not be used for production.

2. System Architecture

2.1 Solution Components

The VR624 solution consists of two core components:

On-Vehicle Router Unit (VR624) and External Antennas

2.2 Deployment

Each vehicle is equipped with one VR624 router in a standard configuration:

Component	Quantity	Description
VR624 Main Unit	1	Mounted in vehicle cabin, DIN-rail or wall-mount
5G/4G Antennas	4/2	External antennas for cellular connectivity
Wi-Fi Antennas	2	External antennas for Wi-Fi coverage
GNSS Antenna	1	GPS antenna for positioning
SIM Cards	1	Dual Nano-SIM with failover support
Power Supply	1	DC 9-48V input with protection
Ethernet Connections	Up to 4	M12 X-code connectors for in-vehicle devices

3. Hardware Specifications

3.1 Hardware Overview

Parameter	Specifications
Processor	1.3 GHz industrial-grade CPU
Memory	512 MB DDR4
Storage	8 GB eMMC
Cellular Module	5G NR NSA/SA, downlink up to 3.4 Gbps
SIM Card Slots	1× drawer-type dual Nano-SIM slot + 1× eSIM (optional)
Ethernet Ports	4× M12 X-code 8-pin female, 10/100/1000 Mbps

Parameter	Specifications
Serial Port	1× RS232 with 15 KV ESD protection
I/O Interfaces	2× Input (ignition detection / low battery)
Wi-Fi	IEEE802.11 ax/ac/a/b/g/n, Wi-Fi 6, 2.4GHz + 5GHz, 3000 Mbps
Wi-Fi Security	WPA3-Enterprise, WPA2-Enterprise, WEP/TKIP/AES encryption
Wi-Fi Max Power	5 GHz: 20 dBm, 2.4 GHz: 20 dBm
Wi-Fi Coverage	50 meters line of sight (actual distance depends on environment)
GNSS Receiver	GPS, BDS, Galileo, GLONASS, QZSS
Positioning Accuracy	< 1.5 m (CEP50)
Tracking Sensitivity	-162 dBm
Navigation Update Rate	10 Hz
Antenna Interfaces	4× SMA (5G), 2× SMA (4G), 2× RP-SMA (Wi-Fi), 1× SMA (GNSS)
Indicators/LEDs	1× System, 1× Cellular, 1× Signal, 1× 2.4G Wi-Fi, 1× 5G Wi-Fi
Ground Terminal	Supported
Security Chip	TPM 2.0
Real-Time Clock	RTC supported
Power Supply	DC 9-48 V, over-current protection, anti-reverse connection
Power Consumption	Typical operating power (refer to detailed specs)
Reset Button	Reset button for factory restore
Housing	Metal enclosure
Cooling	Fanless passive cooling
Protection Rating	IP30
Operating Temperature	-30°C ~ +70°C
Storage Temperature	-40°C ~ +85°C
Ambient Humidity	5% ~ 95% (non-condensing)
Dimensions	177 × 144.25 × 43 mm

Parameter	Specifications
Weight	810 g
Installation	Wall-mounting
MTBF	MIL-HDBK-217F N2, GM @ 20°C: 17.8 years MIL-HDBK-217F N2, GM @ 40°C: 12.5 years

3.2 Interface Overview

M12 Ethernet Ports (×4)

- M12 X-code 8-pin female connectors
- 10/100/1000 Mbps auto-negotiation
- Vibration-resistant design for mobile environments

Serial Port (×1)

- RS232 interface with 15 KV ESD protection
- For legacy device integration

I/O Interfaces (×2)

- Input 1: Ignition detection
- Input 2: Low battery warning

3.3 Environmental & Reliability Specifications

Test Item	Standard	Level / Result
ESD (Electrostatic Discharge)	EN61000-4-2, ISO16750-2:4.7	Level 3
Radiated Immunity	EN61000-4-3, ISO16750-2: 4.6	Level 3
EFT (Electrical Fast Transient)	EN61000-4-4, ISO16750-2: 4.3	Level 3
Surge	EN61000-4-5, ISO16750-2: 4.4	Level 3
Conducted Immunity	EN61000-4-6, ISO16750-2: 4.5	Level 3
Oscillatory Waves	EN61000-4-12, ISO16750-2: 4.8	Level 3
Power Frequency Magnetic Field	EN61000-4-8, ISO16750-3: 4.1.2	Horizontal/Vertical 400 A/m (≥ Level 2)

Test Item	Standard	Level / Result
Shock	IEC60068-2-27, ISO16750-3: 4.1.1	Compliant
Vibration	IEC60068-2-6, ISO16750-3: 4.1.2	Compliant
Free Fall	IEC60068-2-32, ISO16750-3: 4.1.3	Compliant
Rail Standard EMC	EN50155, EN50121	Compliant
Fire Prevention	EN45545	Compliant

3.4 Key Hardware Advantages

Advantage	Description
M12 X-Coded Connectors	Vibration-resistant, corrosion-resistant, ensuring reliable wired connections in harsh mobile environments, flow ITxPT standards
Wide-Temperature Design	-30°C to +70°C operating range for extreme weather conditions
TPM 2.0 Security	Hardware-level encryption for enterprise-grade security
Dual-SIM Failover	Automatic SIM switching for uninterrupted connectivity
Multi-Constellation GNSS	GPS + BDS + Galileo + GLONASS + QZSS for global positioning
Fanless Cooling	No moving parts for enhanced reliability and maintenance-free operation

4. Software Features

4.1 Network Protocols

Protocol Category	Supported Protocols
Network Access	APN, VPDN
Authentication	CHAP, PAP
LAN Protocol	ARP, Ethernet
WAN Protocol	Static IP, DHCP, PPPoE
Routing	Static routing, BGP, OSPF
TCP/IP Stack	TCP, UDP, IPv4, IPv6, ICMP, NTP, DNS, HTTP, HTTPS, SSL/TLS, ARP, VRRP, PPP, PPPoE, SSH
DHCP Services	DHCP Server, DHCP Client
DNS Services	DNS proxy, DNS filtering, custom DNS servers
DDNS	Dynamic DNS, multi-provider support

4.2 Network Security

Security Feature	Description
Firewall	Stateful packet inspection, inbound/outbound rules, port forwarding, NAT (SNAT/DNAT) MAC filtering, domain filtering, LAN firewall
Access Control	802.1X authentication, MAC/IP/port/protocol-based access control
Policy Routing	Source/destination-based routing, domain-based routing, forced routing
VPN Support	IPsec VPN, L2TP VPN, OpenVPN, WireGuard VPN
TPM 2.0	Trusted Platform Module for hardware-level encryption
Traffic Shaping (QoS)	Interface-level bandwidth control, queue-based traffic management

4.3 Link Management & Backup

Interface Backup

- WAN/cellular link failover
- Multiple backup modes: primary/standby, load balancing
- Detection methods: ICMP

Link Monitoring

- Real-time link status, latency, jitter, packet loss
- Historical link quality trends
- Automated health checks
- Instant alert notifications

Dual-SIM Failover

- Automatic SIM switching based on: Flow threshold, Dial failure, Abnormal time periods, Detection failure
- Configurable failover triggers
- Intelligent fallback mechanism

Link Detection

- Heartbeat packet detection
- Auto-redial on disconnection
- Embedded watchdog for automatic recovery

4.4 Wi-Fi Features

Wi-Fi Operating Modes

- AP Mode: Access Point for client connections
- Client Mode: Connect to external Wi-Fi networks

Wi-Fi Security

- Open system, shared key authentication
- WPA-PSK/WPA2-PSK/WPA3-PSK/WPA/WPA2/WPA3 certification
- TKIP/AES encryption

Wi-Fi Portal

- Built-in portal server
- Multiple authentication methods: username/password, WeChat
- Whitelist and time-based exempt authentication

4.5 DTU Functions

Transparent Transmission

- TCP/UDP transparent transmission mode
- Modbus RTU to Modbus TCP bridge conversion

4.6 System Management

Dashboard

- Device information overview
- Interface status display
- Refined traffic statistics

Link Monitoring

- Link delay, jitter, packet loss, throughput

Logs & Alarms

- System logs
- Diagnostic logs
- Device events
- Configurable alarm rules
- Email alarm notifications

Network Diagnostics

- Ping: Network connectivity testing
- Traceroute: Route tracing
- Packet capture: Network data analysis
- Iperf: Network performance testing

Configuration Management

- Configuration import/export
- Backup and restore
- Firmware upgrades (Web/Cloud/Automatic)
- Module firmware upgrades

Remote Access

- Web management interface
- CLI (Command Line Interface)
- Remote Web/CLI acces

4.7 Positioning Service

GNSS Positioning

- GNSS IP forwarding, supporting NMEA RMC, GSA, GAA, GSV
- GNSS Serial port forwarding, supporting NMEA RMC, GSA, GAA, GSV
- Reporting interval, configurable

Location-Based Services

- Real-time vehicle tracking
- Route optimization
- Geofencing alerts
- Historical route playback

5. Cloud Management & Fleet Operations

5.1 DeviceLive Platform Integration

VR624 integrates seamlessly with InHand Networks DeviceLive IoT device management platform, enabling centralized management of thousands of distributed in-vehicle routers.

5.2 Cloud Management Capabilities

Capability	Description
Real-Time Monitoring	Live device status, signal strength, data usage visualization
Remote Diagnostics	Remote troubleshooting without physical access

Capability	Description
Batch Management	Configure multiple devices simultaneously
Fleet Monitoring	Track entire fleet from single dashboard
Remote Maintenance	Reduce truck rolls with remote configuration and troubleshooting
OTA Updates	Over-the-air firmware and configuration updates
Alert Management	Instant notifications for critical events
Location Tracking	Real-time vehicle positioning and route tracking

6. Ordering Guide

6.1 Model Code Structure

Model	Description	Region
VR624-GLNR	**5G NR NSA** n1/2/3/5/7/8/12/13/14/18/20/25/26/28/29/30/38/40/41/48/66/70/71/75/76/77/78/79 **5G NR SA** n1/2/3/5/7/8/12/13/14/18/20/25/26/28/29/30/38/40/41/48/66/70/71/75/76/77/78/79 **LTE FDD** B1/2/3/4/5/7/8/12/13/14/17/18/19/20/25/26/28/29/30/32/66/71 **LTE TDD** B34/38/39/40/41/42/43/48 **LTE LAA** B46 **WCDMA** B1/2/4/5/8/19	**Global**
VR624-GLC6	**LTE-FDD** B1/2/3/4/5/7/8/12/13/14/17/18/19/20/25/26/28/29/30/32/66/71 **LTE-TDD** B34/38/39/40/41/42/43/46(LAA)/48(CBRS) **WCDMA** B1/2/3/4/5/6/8/19	**Global**

6.2 Package Contents

Standard Package

- VR624 5G Vehicle Router × 1
- Mounting kit × 1
- Quick Start Guide × 1

Optional Accessories

- M12 to RJ45 Ethernt cable
- Antenna kit (Cellular + Wi-Fi + GNSS)
- Power cable

- DIN-rail mounting bracket

6.3 Certifications*

Certification Type
CE
E-Mark (ECE R10, R118)
FCC
PTCRB
ITxPT

The product is currently undergoing certification.

7. Installation & Deployment

7.1 Mounting Options

Wall Mounting

- Standard wall-mounting bracket included
- Screw mounting for secure installation

DIN-Rail Mounting (Optional)

- Optional DIN-rail kit for industrial environments
- Tool-less installation and removal

7.2 Installation Procedure

Step	Task
1	Mount VR624 unit in vehicle cabin
2	Connect external antennas (cellular/Wi-Fi/GNSS)
3	Insert SIM cards (dual-SIM configuration)
4	Connect power supply (DC 9-48V)
5	Connect Ethernet devices (up to 4)

Step	Task
6	Connect serial/I/O cables (if required)
7	Power on and verify system status

Total installation time: Approximately 20-30 minutes

7.3 Installation Notes

- Ensure antenna placement for optimal signal reception
- Verify power supply voltage (9-48V DC) before connection
- Ground the device properly for EMC compliance
- Route cables away from heat sources and moving parts
- Confirm SIM card orientation before insertion

8. Technical Support

8.1 Contact Information

- **Technical Support Hotline:** +1 (703) 348-2988
- **Technical Support Email:** support@inhand.com
- **Online Documentation:** https://www.inhand.com/en/resources-center/
- **Cloud Management Platform:** https://device.inhandcloud.com/

8.2 Warranty Information

- Warranty period: 3 Years
- Extended warranty options available

9. About InHand Networks

InHand Networks is a leading IoT solutions provider founded in 2001, dedicated to driving digital transformation across industries and empowering customers to unlock their full potential and achieve accelerated growth.

We specialize in delivering industrial-grade connectivity solutions for diverse sectors, such as enterprise networks, industrial and building IoT, digital energy, smart commerce, and mobility. Our comprehensive product portfolio and services cater to various applications worldwide, including smart manufacturing, smart grid, intelligent transportation, smart retail, and more. With a global footprint spanning over 60 countries, we serve customers in China, the United States, France, Germany, the United Kingdom, Italy, and beyond.

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